

Tahmina Sultana Priya

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RESEARCH INTEREST

Machine Learning for Health, Clinical Phenotyping, Disease Progression and Risk Prediction, Explainable AI, Precision Medicine, Large Language Models, Agentic AI

EDUCATION

Virginia Polytechnic Institute and State University

Virginia, USA

Ph.D. in Computer Science

January 2023 – Present

Supervised by: Dr. Danfeng (Daphne) Yao

Expected Graduation: December 2026

Dissertation: Algorithm-driven Precise and Explainable Prognosis for Complex Metabolic Disorders

Virginia Polytechnic Institute and State University

Virginia, USA

M.Sc. in Computer Science

GPA: 4.00/4.00 | January 2023 – May 2025

Khulna University of Engineering & Technology

Khulna, Bangladesh

B.Sc. in Computer Science and Engineering

CGPA: 3.77/4.00 | Rank: 7th/134 | March 2016 – March 2020

PUBLICATIONS & PRESENTATIONS

1. **Tahmina Sultana Priya**, Huihuang Yan, Kirk J. Wangenstein, Stephen Wu, Anthony C. Luehrs, Filippo Pinto e Vairo, Fan Leng, Andres J. Acosta, Robert A. Vierkant, Alina M. Allen, Konstantinos N. Lazaridis, Eric W. Klee, Danfeng (Daphne) Yao, Shulan Tian "Subtyping Metabolic Dysfunction-Associated Steatotic Liver Disease using Electronic Health Record-Linked Genomic Cohorts Reveals Diverse Etiologies and Progression" *Published in Nature Communications*, 2026.
2. **Tahmina Sultana Priya**, Kirk J. Wangenstein, Eric W. Klee, Shulan Tian, Danfeng (Daphne) Yao "Explainable and subgroup-specific machine learning models to predict rapid fibrosis progression in MASLD" *Under Submission*, 2026.
3. **Tahmina Sultana Priya**, Shulan Tian, Danfeng (Daphne) Yao "Differential Subgroup Reasoning and Risk-Aware Explainability for Metabolic Disorders" *Document in preparation*, 2026.
4. Mohimenul Karim, **Tahmina Sultana Priya**, Patricia C. Scherer, Justin F. Rousseau, Xin Zhang, Chang Lu, Jeffrey D. Shahidullah, D. Jeffrey Newport, Karen Dineen Wagner, Charles B. Nemeroff, Danfeng (Daphne) Yao "Precision Characterization of Adverse Childhood Experience, Family History, and Recovery in Trauma-exposed Children and Adolescents" *Document in preparation*, 2026.
5. Huihuang Yan, Andres J. Acosta, Srushti Gangireddy, **Tahmina Sultana Priya**, Josey Sorenson, Tara L. Haynes, Anthony Batzler, Anthony C. Luehrs, Hugues P. Sicotte, Wenan Chen, Shannon McDonnell, Sebastian Armasu, Robert A. Vierkant, Jan Egan, Kirk J. Wangenstein, Daniel Schaid, Konstantinos N. Lazaridis, Wei-Qi Wei, Shulan Tian, Eric W. Klee "Subtype-specific comorbidity risks and genetic etiologies in obesity" *Document in preparation*, 2026.
6. Tanmoy Sarkar Pias, Wenjia Song, **Tahmina Sultana Priya**, Sahil Dudani, Randy Marchany, Brad Tilley, Lingxiang Wang, Danfeng (Daphne) Yao "On the Readiness of Enterprise-scale Log Analysis Solutions Against Advanced Persistent Threats" *Published in IEEE Secure Development (SecDev) Conference*, October 10th, 2025, Indianapolis, United States.
7. **Tahmina Sultana Priya**, Fan Leng, Anthony C. Luehrs, Eric W. Klee, Alina M. Allen, Konstantinos N. Lazaridis, Danfeng (Daphne) Yao, Shulan Tian "Deep Phenotyping of Non-Alcoholic Fatty Liver Disease Patients with Genetic Factors for Insights into the Complex Disease." *Published in Machine Learning for Health (ML4H)*, December 10th, 2023, New Orleans, United States.
8. **Tahmina Sultana Priya**, Huihuang Yan, Konstantinos N. Lazaridis, Eric W. Klee, Danfeng (Daphne) Yao, Shulan Tian, "Prediction of Rapid Fibrosis Progression in Metabolic Dysfunction-Associated

Steatotic Liver Disease with Interpretable Machine Learning Models” - Presented at 14th **International Conference on Intelligent Biology and Medicine**, Buffalo, NY, USA; Aug 2-5, 2026; **Talk**

9. **Tahmina Sultana Priya**, “Algorithm-driven Precise and Explainable Prognosis for Complex Disorders” - Presented at AHLI 7th **Annual Conference on Health, Inference, and Learning and Summer Camp**, Seattle, WA, USA; June 21-28, 2026; **Poster**
10. **Tahmina Sultana Priya**, Huihuang Yan, Stephen Wu, Filippo Pinto e Vairo, Anthony C. Luehrs, Fan Leng, Robert A. Vierkant, Eric W. Klee, Alina M. Allen, Konstantinos N. Lazaridis, Shulan Tian, Danfeng (Daphne) Yao “Subtyping of metabolic dysfunction-associated steatotic liver disease (MASLD) by real-world integration of electric health records and population genomics data” - Presented at the 32nd Conference on **Intelligent Systems for Molecular Biology (ISMB)**, Montreal, QC, Canada; June 12-16, 2024; **Poster**
11. **Tahmina Sultana Priya**; Fan Leng; Anthony C. Luehrs; Huihuang Yan, Eric W. Klee; Alina M. Allen, Konstantinos N. Lazaridis, Shulan Tian, Danfeng (Daphne) Yao “Latent class analysis-based identification of non-alcoholic fatty liver disease subtypes with distinct metabolic and genetic signatures” - Presented at the 20th Annual Meeting of **MCBIOS**, Emory University, Atlanta, GA, USA; March 22-24, 2024; **Lightning Talk**

RESEARCH EXPERIENCE

Virginia Polytechnic Institute and State University

Virginia, USA

Graduate Research Assistant, Department of Computer Science

May 2023 - Present

- Conduct healthcare data analytics using electronic health records (EHR), clinical, and genomic datasets to generate insights supporting disease characterization, patient outcomes, and precision healthcare research.
- Extract, clean, harmonize, and integrate large-scale healthcare datasets from EHR and genomic sources to identify clinically meaningful patient subgroups and characterize disease patterns.
- Develop and evaluate predictive and statistical models for disease prognosis, risk stratification, and healthcare outcome analysis in complex metabolic disorders.
- Develop explainable and subgroup-specific predictive models to identify clinical factors associated with disease progression and support interpretable healthcare decision-making.
- Design a multi-agent LLM framework that generates clinically grounded, subgroup-specific explanations for statistically-assigned patient subtypes, integrating retrieval-augmented evidence from cohort data and biomedical literature.
- Develop evaluation methods to assess the faithfulness, factual grounding, and reliability of LLM-generated clinical explanations, including consistency checks against empirical patient data and independent feature-attribution methods.

Khulna University of Engineering & Technology

Khulna, Bangladesh

Undergraduate Thesis (Supervised by: Dr. Muhammad Aminul Haque Akhand)

June 2019-Mar 2020

- Addressed the NP-Hard problem of the Nurse Rostering Problem (NRP) with a focus on creating both valid and optimal solutions using a Modified Particle Swarm Optimization approach
- Achieved better results compared to various other techniques, leveraging the INRC-II dataset

WORK EXPERIENCE

Mayo Clinic

Rochester, MN, USA

Research Intern – Bioinformatics, Division of Computational Biology

(Supervised by: Dr. Shulan Tian)

May 2025 – August 2025

January 2024 – August 2024

- Analyzed electronic health records (EHR), clinical, and genomic datasets to extract meaningful healthcare insights and identify factors associated with disease progression.
- Performed data preprocessing, integration, validation, and statistical analysis of large-scale biomedical datasets.
- Developed predictive machine learning models for healthcare analytics, patient risk assessment, and outcome prediction.

- Identified clinically meaningful disease subgroups using multi-omics data integration and computational phenotyping approaches.
- Conducted survival analysis and drug response analysis to support precision medicine research and clinical decision-making.

Virginia Polytechnic Institute and State University

Graduate Teaching Assistant, Department of Computer Science

Virginia, USA

January 2023 – December 2025

- Helped students grasp core concepts in Data Structures and Algorithms, Computer Organization, and Android Development.
- Provided guidance on programming, system architecture, and mobile app development.
- Developed project solutions and graded assignments.

Daffodil International University

Lecturer, Department of Computer Science and Engineering

Dhaka, Bangladesh

January 2021 – December 2022

- Taught courses including Web Engineering, Data Mining, Machine Learning, Software Engineering, Digital Electronics, Software Project, Numerical Methods, and Data Communication.
- Instructed over 200 students and supervised 5 student project groups.

Vertical Innovations Limited

Senior Software Engineer

System Engineer

Dhaka, Bangladesh

July 2022 – December 2022

August 2020 – June 2022

- Served as a team lead to build scalable web applications using Django and Nuxt.js.
- Developed data analytics dashboards and visualizations using Python.
- Integrated online payment gateways and deployed cloud-based infrastructure.
- Designed IoT-based data collection and monitoring systems and supported platform operations.

KEY SKILLS

ML/DL Models:	XGBoost, Random Forest, LightGBM, Logistic Regression, MLP, CNN, Autoencoder, LSTM, GRU
Large Language Models:	Llama, Deepseek-r1, Qwen, Phi, RAG, Prompt Engineering, Multi-Agent Systems, Zero/Few-shot learning, Reasoning, CoT
XAI & Model Evaluation:	SHAP, Feature Attribution, Model Calibration, Counterfactual Analysis, Dimensionality Reduction (PCA, t-SNE)
Statistics:	ANOVA, t-test, Chi-square, Cox Proportional Hazards
Languages & Frameworks:	Python, R, Scikit-learn, Pandas, C/C++, JAVA, NumPy, Matplotlib , PyTorch, HuggingFace Transformers
Healthcare Data:	Electronic Health Records, Genomic Data, Clinical Notes, ICD Codes, Billing Codes
Databases:	SQL, PostgreSQL, MySQL, Oracle
Tools & Platforms:	Git, VS Code, WEKA

AWARDS & ACHIEVEMENTS

- CHIL and ICIBM Conference Travel Award (2026)
- Pratt Fellowship, Virginia Tech (2024–2025)
- Dean’s Award, KUET (2017–2018, 2019–2020)
- 1st Runner-up, Textile Fest KUET – Fire-fighting Robot Project
- Government Scholarships, Bangladesh (General and Talent Pool)

REVIEWER

IEEE Journal of Biomedical and Health Informatics (JBHI), Machine Learning for Healthcare (MLHC), Machine Learning for Health (ML4H), Nature Scientific Reports